AGA KHAN UNIVERSITY EXAMINATION BOARD

SECONDARY SCHOOL CERTIFICATE

CLASS IX

ANNUAL EXAMINATIONS (THEORY) 2024

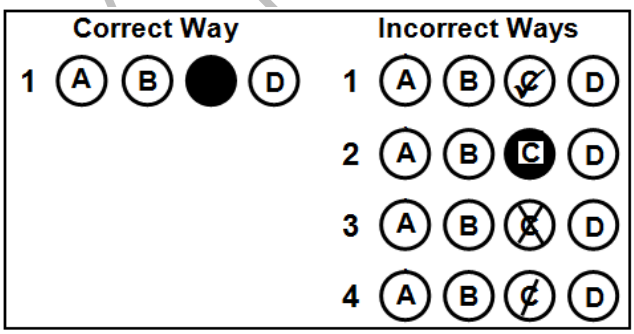
General Mathematics Paper I

Time: 1 hour 20 minutes Marks: 45

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INSTRUCTIONS

1. Read each question carefully.
2. Answer the questions on the separate answer sheet provided. DO NOT write your answers on the question paper.
3. There are 100 answer numbers on the answer sheet. Use answer numbers 1 to 45 only.
4. In each question, there are four choices A, B, C, D. Choose ONE. On the answer grid, black out the circle for your choice with a pencil as shown below.



Candidate's Signature

5. If you want to change your answer, ERASE the first answer completely with a rubber, before blacking out a new circle.
6. DO NOT write anything in the answer grid. The computer only records what is in the circles.
7. A formulae list is provided on page 2. You may refer to it during the paper, if you wish.
8. You may use a simple calculator if you wish.

List of Formulae

Note:

All symbols used in the formulae have their usual meaning.

Business Mathematics

$$\text{Loss }\% = \left(\frac{\text{Loss}}{CP} \times 100\right) \qquad \text{Profit }\% = \left(\frac{\text{Profit}}{CP} \times 100\right) \qquad SP = CP \times \left(\frac{100 + \text{profit}\,\%}{100}\right)$$

$$SP = CP \times \left(\frac{100 - \text{loss}\%}{100}\right) \qquad \text{Discount }\% = \frac{\text{Discount}}{MP} \times 100$$

Sets and Functions

$$(A \cup B)^c = A^c \cap B^c \qquad (A \cap B)^c = A^c \cup B^c$$

Exponents and Logarithms

$$a^m \times a^n = a^{m+n} \qquad a^m \div a^n = a^{m-n} \qquad \frac{a^m}{b^m} = \left(\frac{a}{b}\right)^m$$

$$\left(a^m\right)^n = a^{mn} \qquad (a \times b)^m = a^m \times b^m \qquad a^{\frac{m}{n}} = \sqrt[n]{a^m}$$

$$\log_a (m)^n = n \log_a m \qquad \log_a (m \times n) = \log_a m + \log_a n \qquad \log_a \left(\frac{m}{n}\right) = \log_a m - \log_a n$$

$$\log_a n = \log_b n \times \log_a b \qquad \log_a b = n \Leftrightarrow a^n = b$$

Algebraic Formulae and Applications/ Factorisation

$$(a-b)^2 = a^2 - 2ab + b^2 \qquad (a+b)^2 = a^2 + 2ab + b^2$$

$$a^2 - b^2 = (a+b)(a-b) \qquad 2\left(a^2 + b^2\right) = (a+b)^2 + (a-b)^2$$

$$(a+b)^2 - (a-b)^2 = 4ab \qquad (a+b+c)^2 = a^2 + b^2 + c^2 + 2ab + 2bc + 2ca$$

Linear Graphs

$$1 \text{ mile} = \frac{8}{5}\text{km} \qquad 1 \text{ Hectare} = 2.471 \text{ Acres} \qquad {}^\circ F = \left(\frac{9}{5} \times {}^\circ\text{C}\right) + 32$$

Matrices and Determinants

$$A^{-1} = \frac{1}{|A|} AdjA$$

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1. 250% of 25 is equal to

   A. 122.5
   B. 62.5
   C. 12.25
   D. 6.25

2. The ratio of $2a$ to $\frac{a}{5}$ is equal to

   A. 2:5
   B. 5:2
   C. 10:1
   D. 1:10

3. Irtiza scored 85% marks in a test of 60 marks. If he wants to score 95%, then the difference in the obtained marks would have been

   A. 10
   B. 9
   C. 6
   D. 3

4. A piece of rubber band has an original length of 30 cm. If it is stretched to a length of 35 cm, then the length is increased in the ratio of

   A. 5:6
   B. 7:6
   C. 30:5
   D. 35:5

5. The force of attraction between two objects increases with the product of their masses and decreases with the increase of the square of the distance between them.

   The given statement is an example of

   A. compound proportion.
   B. inverse proportion.
   C. direct proportion.
   D. joint proportion.

6. The *Ushr* due on an income of Rs 100,000 from agricultural products will be

   (**Note:** Rate of *Ushr* is 10% of earning.)

   A. Rs 1,000
   B. Rs 9,000
   C. Rs 10,000
   D. Rs 90,000

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7. Basma saved Rs 800,000 in a year. If the rate of *zakat* is 2.5% per annum, then the amount of *zakat* she will be paying is

   A. Rs 2,000
   B. Rs 20,000
   C. Rs 40,000
   D. Rs 200,000

8. A deceased person's estate, totaling Rs 1,000,000, will be divided among his legal heirs, consisting of two sons and one daughter.

   Based on the given situation, the share of the daughter will be

   (**Note**: The share of a son is two times that of a daughter.)

   A. Rs 200,000
   B. Rs 500,000
   C. Rs 600,000
   D. Rs 800,000

9. The sale price of a book is Rs 1,500. If a seller had purchased for Rs 1,200, then the profit earned on the book by the seller will be

   A. Rs 300
   B. Rs 600
   C. Rs 900
   D. Rs 1,800

10. A gold chain bought for Rs 50,000 is sold after two years at a profit of Rs 9,500. The cost of this chain has increased by

    A. 19%.
    B. 23.5%.
    C. 119%.
    D. 123.5%.

11. The marked price of a laptop is Rs 75,000. After a discount of 15%, the sale price will be

    A. Rs 11,250
    B. Rs 56,250
    C. Rs 63,750
    D. Rs 86,250

12. The tag price of a wrist watch is Rs 15,000. If the sale price is Rs 14,000, then the discount rate will be equal to

    A. 6.66%.
    B. 7.14%.
    C. 14.0%.
    D. 15.0%.

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13. In a joint business, Saif invested 120% as much as Karim and Javed invested 80% as much as Karim.

    If Karim invested Rs 500,000, then the total investment was

    A. Rs 400,000
    B. Rs 500,000
    C. Rs 600,000
    D. Rs 1500,000

14. If $X = \{1, 3, 5\}$, $Y = \{1, 5\}$, then which of the given options is TRUE?

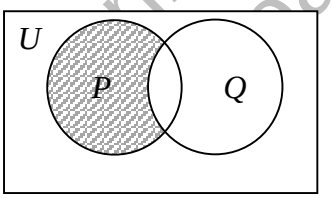
    I. $X \cup Y = X$
    II. $X \cap Y = X$
    III. $X \cup Y = Y$
    IV. $X \cap Y = Y$

    A. I and II
    B. II and III
    C. I and IV
    D. III and IV

15. For three non-empty sets *X*, *Y* and *Z*, $X \cap (Y \cup Z) = (Y \cup Z) \cap X$. This property is called

    A. commutative property of intersection.
    B. associative property of intersection.
    C. commutative property of union.
    D. associative property of union.

16. The shaded region in the given Venn diagram represents



    A. P'.
    B. Q'.
    C. $P - Q$.
    D. $Q - P$.

17. The domain of the binary relation $r = \{(d, 11), (e, 22), (f, 33)\}$ is

    A. {11, 22, 33}.
    B. {*d*, 11}.
    C. {*d*, *e*, *f* }.
    D. {*d*, *e*, *f*, 11, 22, 33}.

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18. The $M = \{1, 2\}$ and $N = \{0\}$, then which of the given options is a function M to N?

A. $\{(1, 0)\}$
B. $\{(1, 0), (2, 0)\}$
C. $\{(1, 0), (0, 2)\}$
D. $\{(1, 0), (0, 2), (2, 0\}$

19. On simplification of $\left(\frac{x^2}{x}\right)^{-2}$, we get

A. $\frac{1}{x^2}$.
B. $\frac{1}{x^5}$.
C. $\frac{1}{x^4}$.
D. $\frac{1}{x}$.

20. The radicand of the expression $\sqrt[3]{3x^2}$ is

A. $\frac{1}{3}$
B. 3
C. $3x$
D. $3x^2$

21. The exponential form of the expression $\sqrt[4]{\frac{a^3}{b^3}}$ will be

A. $\left(\frac{a}{b}\right)^{\frac{3}{4}}$
B. $\left(\frac{a}{b}\right)^{\frac{4}{3}}$
C. $(a-b)^{\frac{3}{4}}$
D. $(a-b)^{\frac{4}{3}}$

22. If $a \times 10^{-3} = 0.0034$, then the value of a will be

A. -3.4
B. -0.34
C. 0.34
D. 3.4

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23. If the expression $(a+b)^x = y$, then in form of logarithm, the expression will be

A. $\log_{(a+b)} y = x.$
B. $\log_y x = a + b.$
C. $\log_x y = a + b.$
D. $\log_y (a+b) = x.$

24. In expanded form, $\log\left(\frac{w}{xy}\right)$ can be expressed as

A. $\log w \div (\log x \times \log y).$
B. $\log w - \log x - \log y.$
C. $\log w - \log x + \log y.$
D. $\log w \div \log xy.$

25. If $\log_2 3 = 1.5849$, then the value of $\log_2 6$ will be

A. 3.1698
B. 2.5849
C. 1.5849
D. 0.7924

26. Which of the following algebraic expressions represents a polynomial?

A. $x^2 - \sqrt{x} + 2$
B. $x^2 - x + \sqrt{2}$
C. $x^2 + \frac{1}{x} + \sqrt{2}$
D. $x^{\frac{1}{2}} - x + \sqrt{2}$

27. The expression $\frac{x^2}{x-1} \div \frac{x^2}{x+1}$ can be expressed in the simplest form as

A. 1
B. $x^2 - 1$
C. $\frac{x+1}{x-1}$
D. $\frac{x-1}{x+1}$

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28. If $x = 0,\ y = 1$ and $z = -1$, then the value of the rational expression $\frac{y + z^2 - x}{y - z}$ will be

A. 0
B. $\frac{1}{2}$
C. 1
D. 2

29. If the value of $a + b = 16$ and $2ab = 126,$ then the value of $a^2 + b^2$ will be equal to

A. 110
B. 130
C. 142
D. 256

30. The simplified form of $2\sqrt{x} + \sqrt{x^3}$ will be

A. $3x\sqrt{x}$.
B. $2x\sqrt{x}$.
C. $3x + \sqrt{x}$.
D. $(2 + x)\sqrt{x}$.

31. The rationalised form of the surd $\frac{1}{2 - \sqrt{3}}$ will be

A. $\frac{1}{2 + \sqrt{3}}$
B. $\frac{2 + \sqrt{3}}{2 - \sqrt{3}}$
C. $2 + \sqrt{3}$
D. $-2 + \sqrt{3}$

32. The expression $ax + bx + a + b$ is equal to

A. $(a + b)x.$
B. $(a + b + 1)x.$
C. $(a + b)(x + 1).$
D. $(x + a)(x + b).$

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33. The factorised form of $9-(1-x)^2$ will be

A. $(4+x)(4-x)$.
B. $(4+x)(2-x)$.
C. $(2+x)(2-x)$.
D. $(2+x)(4-x)$.

34. The product of factors of the expression $1-10x+25x^2$ is

A. $(x-5)(x+5)$.
B. $(x-5)(x-5)$.
C. $(1-5x)(1+5x)$.
D. $(1-5x)(1-5x)$.

35. On dividing $2x^3+3x^2-3x$ by $x-1$, the remainder is

A. -3
B. -2
C. 2
D. 3

36. One of the zeros of the polynomial of $(x-1)(x+2)-4$ will be

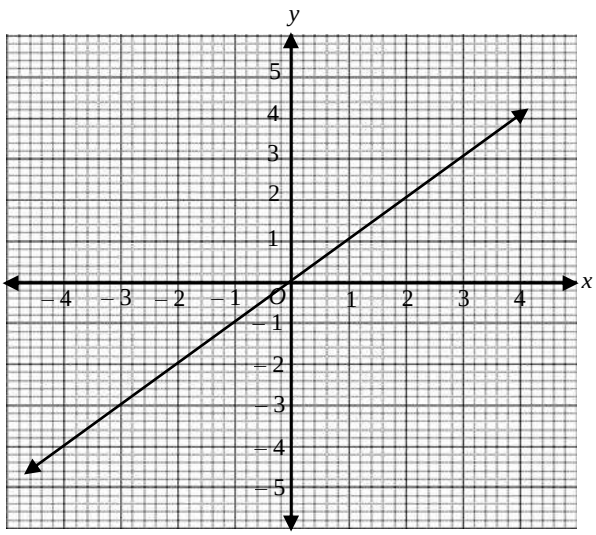
A. -2
B. -1
C. 1
D. 2

37. The point $(-1,2)$ lies in the

A. I quadrant.
B. II quadrant.
C. III quadrant.
D. IV quadrant.

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38. In the given graph, the equation of the given line will be



A. $y = -x$
B. $y = x$
C. $y = x - 1$
D. $y = x + 1$

39. Which of the given points lies on the line $y = x + 1$?

A. $(2, 3)$
B. $(3, 2)$
C. $(1, 1)$
D. $(1, 3)$

40. An example of a row matrix is

A. $\begin{bmatrix} 0 \\ 0 \end{bmatrix}$.
B. $\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$.
C. $\begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$.
D. $\begin{bmatrix} 0 & 1 & 0 \end{bmatrix}$.

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41. If A is non-zero matrix of order 2×2, then the sum of a matrix A and its additive inverse will be

A. $-2A$.
B. $-A$.
C. A.
D. a null matrix.

42. If $E = \begin{bmatrix} 1 & 2 \\ 2 & 0 \end{bmatrix}$, then $3E$ will be equal to

A. $\begin{bmatrix} 3 & 6 \\ 2 & 0 \end{bmatrix}$.
B. $\begin{bmatrix} 3 & 2 \\ 6 & 0 \end{bmatrix}$.
C. $\begin{bmatrix} 3 & 6 \\ 6 & 0 \end{bmatrix}$.
D. $\begin{bmatrix} 3 & 6 \\ 6 & 6 \end{bmatrix}$.

43. If a and b are positive real numbers, then the determinant of the matrix $\begin{bmatrix} a & 1 \\ b & 1 \end{bmatrix}$ will be

A. $a+b$
B. $a-b$
C. $ab-1$
D. $1-ab$

44. If $\begin{bmatrix} 2 & 1 \\ x & 8 \end{bmatrix}$ is a singular matrix, then the value of x will be

A. -16
B. -4
C. 4
D. 16

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45. The multiplicative inverse exists for the matrix

A. $\begin{bmatrix} 1 & 1 \\ 2 & 2 \end{bmatrix}$.

B. $\begin{bmatrix} 0 & 1 \\ 2 & 0 \end{bmatrix}$.

C. $\begin{bmatrix} 1 & 0 \\ 2 & 0 \end{bmatrix}$.

D. $\begin{bmatrix} 1 & 2 \\ 0 & 0 \end{bmatrix}$.

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END OF PAPER

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